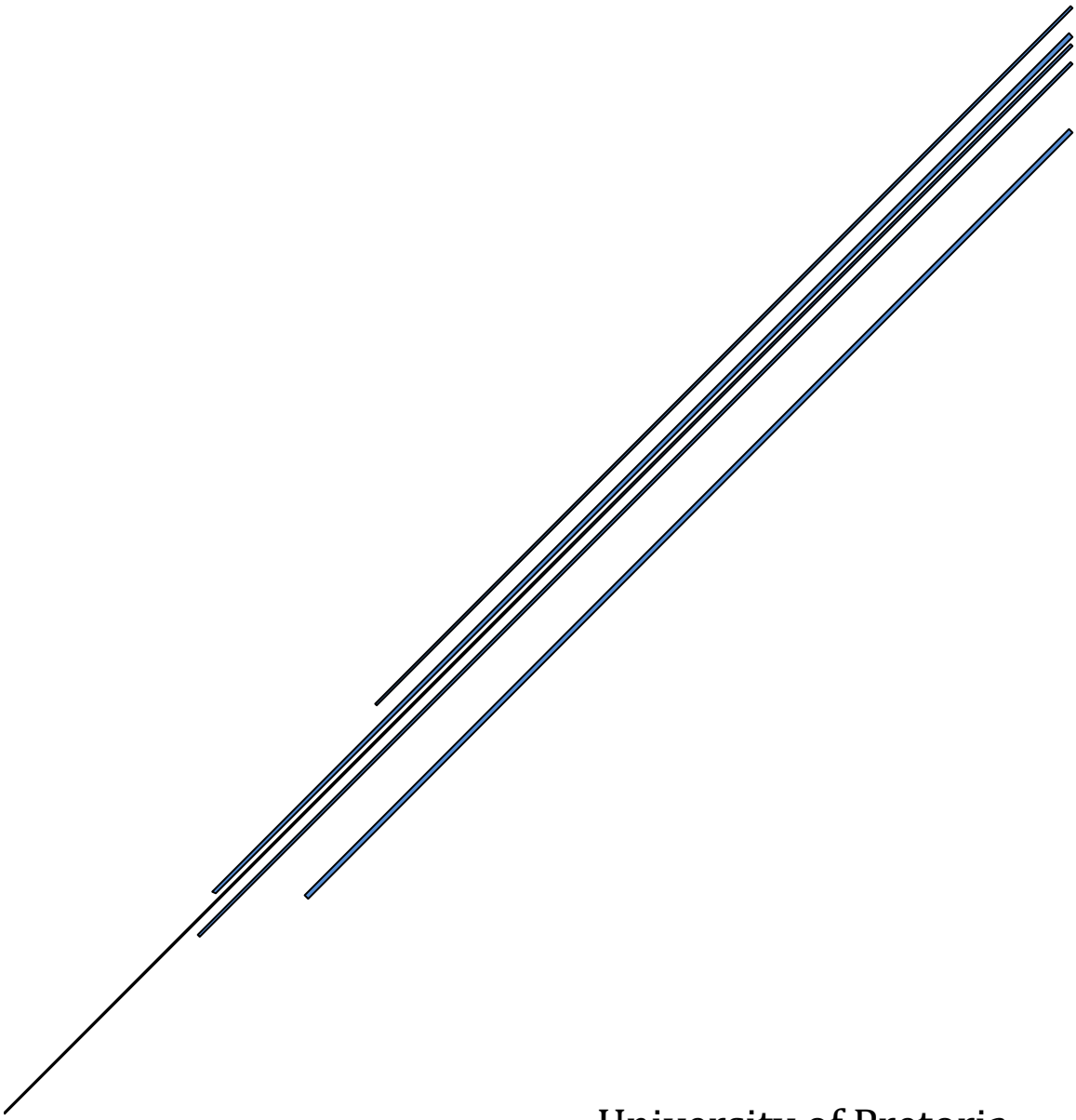
BATCH DISTILLATION EXPERIMENT



University of Pretoria
CLB 321

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Simple Batch Distillation of isopropyl alcohol-water

1. Introduction

In this experiment, you will perform a batch distillation to separate a mixture of isopropyl alcohol (IPA) and water. You will collect and analyse the distillate fractions to determine the concentration of IPA before and after distillation. In parallel, you will also batch distil ethanol water.

2. Equipment and Materials Needed

2.1. Batch Distillation Unit:

- Still pot (round-bottom flask)
- Heating mantle
- Condenser (Liebig)
- Receiver flask or collection vessel
- Boiling chips

2.2. Chemicals:

- Isopropyl alcohol (IPA)
- Distilled water
- Safety Equipment:
- Safety goggles
- Lab coat
- Gloves

2.3. Refractometer:

- Refractometer for measuring the refractive index.

2.4. Polytopes:

- Predetermined polytopes with marked levels (e.g., 10 mL each) for collecting known volumes of distillate.

3. Calibration Procedure for the Refractometer

3.1. Prepare Calibration Solutions:

Create a series of known concentrations of IPA and distilled water (e.g., 10%, 30%, 50%, 70%, and 90% IPA by volume).

3.2. Measure Refractive Indices:

- Use the refractometer to measure the refractive index of each known concentration.
- Record the refractive index for each concentration.

3.3. Generate Calibration Graph/Table:

- Plot the refractive indices against the known concentrations to create a calibration graph.
- Alternatively, tabulate the refractive indices with their corresponding concentrations to form a reference table.

4. Experimental Procedure

4.1. Setup

a. Assemble the distillation apparatus. Connect the still pot (round-bottom flask) to the condenser and ensure the cooling water is properly connected.
b. Fill the still pot with a known mixture of IPA and water (e.g., 50% IPA and 50% water by volume).
c. Add boiling chips to the mixture to prevent bumping.

4.2. Initial Heating

a. Turn the heating mantle to its maximum setting and monitor the flask closely.
b. Once bubbles start to form, indicating the mixture has reached its bubble point, reduce the heating power to just below half of its maximum setting to maintain a steady, controlled boil.

4.3. Distillation

a. As the mixture heats up, the more volatile component (isopropyl alcohol) will preferential congregate in the vapour phase, resulting in the vapour phase becoming richer in IPA compared to the liquid phase.
b. The vapour will travel through the condenser, where it will cool and condense into the liquid phase, then collect in the receiver flask.
c. Monitor the process and adjust the heating if necessary to maintain a steady boil.

4.4. Collecting Fractions

a. Use predetermined polytopes with marked levels (e.g., 10 mL each). As each polytope reaches its marked level, switch it out for the next one.
b. Label each polytope to keep track of the order in which they were collected.

5. Determining Concentration Before & After Distillation

a. Measure the refractive index of the initial mixture (before distillation) using the refractometer.
b. Refer to the calibration graph/table to determine the initial concentration of IPA in the mixture.
c. Do the same for the waste liquid remaining after distillation.

6. Determining Concentration of Distillate

a. Measure the refractive index of each collected fraction using the refractometer.
b. Refer to the calibration graph/table to determine the concentration of IPA in each fraction.
c. Measure the refractive index of the remaining liquid in the still pot after distillation to determine its final concentration.

7. Analysis and Conclusions

7.1. Theoretical Framework

Use the Rayleigh equation to analyse the total amounts of products collected in batch distillation:

$$F = W_{final} + D_{total}$$

$$Fx_F = x_{final}W_{final} + x_{D,avg}D_{total}$$

$$ln\left(\frac{W_{final}}{F}\right) = \int_{x_F}^{x_{final}} \frac{dx_w}{x_D - x_w}$$

7.2. Equilibrium data

Find or simulate equilibrium data to aid in your analysis of the distillation.

7.3. Practical Observations

Compare the experimental data with theoretical predictions. Discuss any discrepancies and potential sources of error.

7.4. Ethanol-water

At your discretion, you may modify/take as is, these instructions for the batch distillation of the ethanol-water system.

8. Conclusions

Draw conclusions based on the collected data, the analysis performed using the Rayleigh equation, and the equilibrium relationships. Discuss the efficiency of the separation process, the purity of the collected fractions, and the practical implications of the results. Write a comprehensive report that adheres to the departmental guidelines on report writing (https://www.up.ac.za/media/shared/Legacy/sitefiles/guide_to_writing_techreports.pdf).